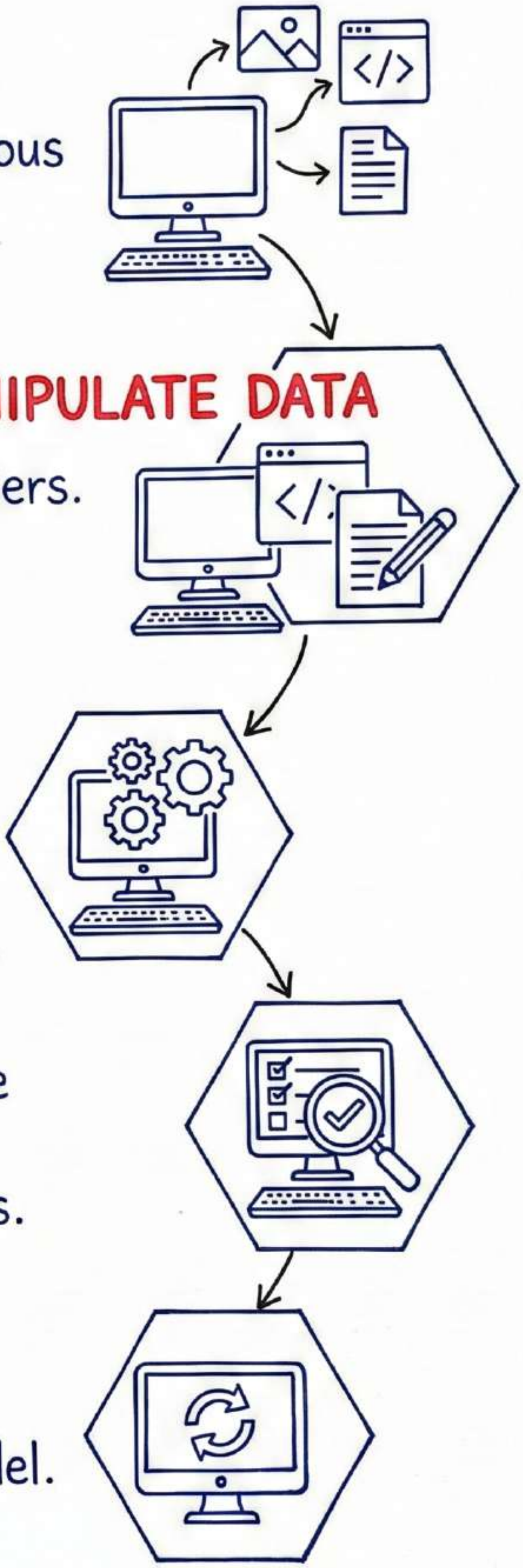
# MACHINE LEARNING WORKFLOW



## 1. GET DATA

- Collect raw data from various sources.
- Important first step.

## 2. CLEAN, PREPARE & MANIPULATE DATA

- Handle missing values, outliers.
- Format data for modeling.
- Crucial for accuracy.

## 3. TRAIN MODEL

- Feed data to algorithms.
- Model learns patterns.

**Exam Tip:** This is an iterative process!

## 4. TEST DATA

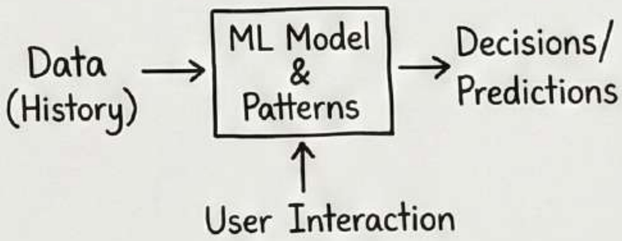
- Evaluate model performance on unseen data.
- Check accuracy and metrics.

## 5. IMPROVE

- Tune hyperparameters.
- Iterate and refine the model.

# 1. Machine Learning (ML)

- Branch of Artificial Intelligence (AI).
- Enables computers to learn from data and make decisions or predictions.
- Identify patterns automatically, without being explicitly programmed.
- Learning is based on historical data and continuous user interaction.
- Common applications: search engines, social media feeds, recommendation and ranking systems.




# 2. Why Ethical Concerns Arise in ML

- ML systems learn from human-generated data.
- This data may contain: bias, discrimination, unfair social patterns.
- Decisions can affect: individuals, groups, society at large.

## Core ethical evaluation questions:


- Is the target system ethical?
- Is the development process ethical?
- Is the data source ethical?
- Is the data itself ethical?
- Is the impact and outcome ethical?

Exam Tip:
Remember ML reflects human data, good or bad!

# 3. Privacy and Surveillance

- Modern digital life leads to constant data collection.
- ML systems rely on personal and personally identifiable information (PII).
- Large-scale data analysis enables continuous surveillance of individuals.

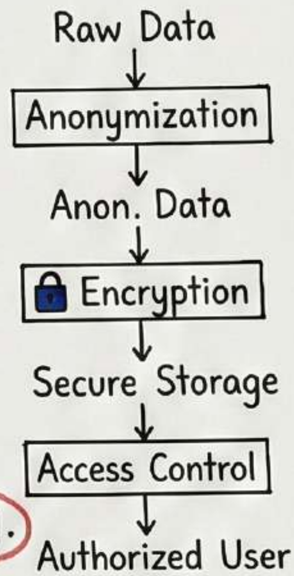
## Aspects of Privacy:

- Right to be left alone.
- Information privacy.
- Control over personal information.
- Right to secrecy.
- **Ethical concern**: Users often lack awareness and control over their data.

Exam Tip: Privacy is about control!

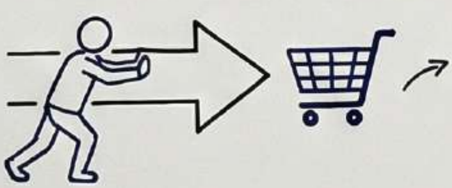
# 4. Privacy-Preserving Techniques

- To reduce risks, ML systems apply:
  - Anonymization: removing identifying attributes.
  - Access control: limiting data access to authorized individuals.
  - Encryption: protecting data during storage and transmission.
- These help but do not eliminate ethical responsibility.




# 5. Manipulation of Human Behavior

- Collected data used to influence or manipulate behavior.
- ML designs personalized nudges based on: user interests, browsing patterns, psychological traits.
- Leads users to: make unintended purchases, subscribe to services, adopt opinions.
- **Ethical issue**: Manipulation reduces user autonomy and informed decision-making.



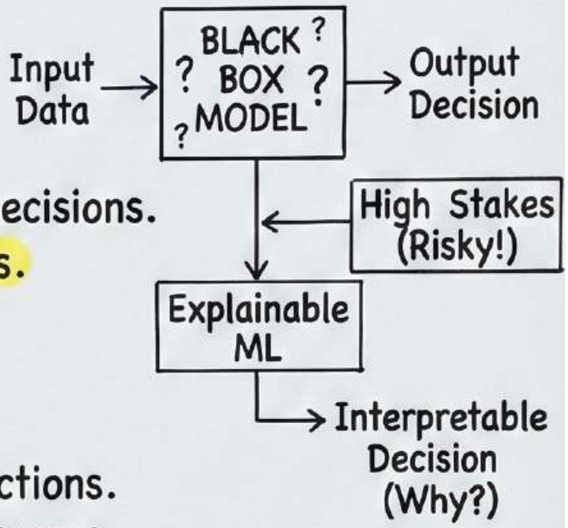
Memory Trick:
ML Nudges → Less Autonomy

## 6. Bias in Decision-Making Systems

- Bias occurs when ML systems produce systematically unfair outcomes.
- Types of bias:
  - Cognitive bias: human prejudice influencing system design.
  - Statistical bias: imbalanced or unrepresentative datasets.
  - Historical bias: past discrimination reflected in training data.
- Bias can lead to discrimination against specific groups.

## 7. Examples of Bias in ML

- **Amazon Recruiting Tool:**
  - Trained on historical hiring data.
  - Dataset was male-dominated.
  - Resulted in gender-biased recruitment decisions.
  - Example of historical and statistical bias.
- **Healthcare Risk Algorithm:**
  - Trained on large-scale patient data.
  - Used biased proxy indicators.
  - Produced racial bias in healthcare predictions.
  - Example of historical bias affecting outcomes.




## 8. Reducing Bias in ML Systems

- Complete elimination of data bias is not fully possible.
- Algorithmic bias can be reduced through:
  - better dataset design, fairness-aware algorithms
- Removing sensitive attributes (race, sex) alone may not solve bias.
- ML interpretability helps:
  - understand decision logic, detect unfair patterns
  - improve accountability

Exam Tip:
Black-box is risky in high-stakes fields!

## 9. Opacity of AI Systems (Black-Box Problem)

- Many ML models (e.g., deep neural networks, random forests) act as black boxes.
- Even developers may not fully understand how decisions are made.
- This creates problems in high-stakes systems such as:
  - healthcare, finance, law enforcement
- Ethical solution: Use explainable and interpretable ML.
- Improves: transparency, fairness, trust, accountability

# CAMBRIDGE ANALYTICA SCANDAL

(Data-Driven Behavioral Manipulation in ML Systems)

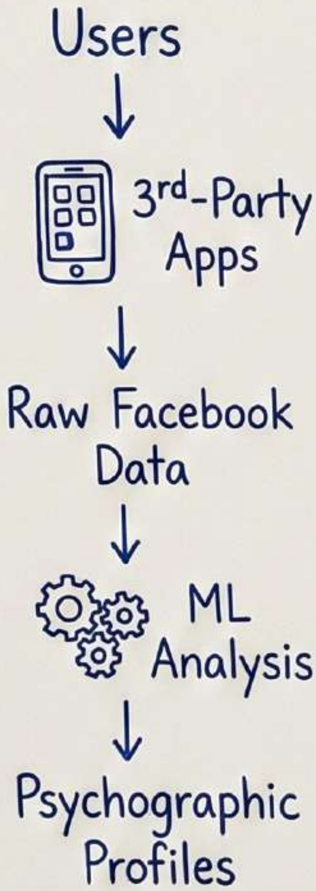
## Background

- Cambridge Analytica was a British political consulting firm [cite: 5].
- It collected massive amounts of **Facebook user data** [cite: 6].
- Data was collected **without informed consent** of users [cite: 7].
- The data was used to influence **political opinions** and **voting behavior** [cite: 8].

Exam Tip: Key is LACK of consent!

## Nature of Data Collected

- Personal Facebook data including: likes [cite: 11], shares [cite: 12], posts [cite: 13], interactions [cite: 14].
- Data was harvested via **third-party apps** [cite: 15].
- Users were largely **unaware** that their data was being collected and reused [cite: 16].




## Use of Machine Learning

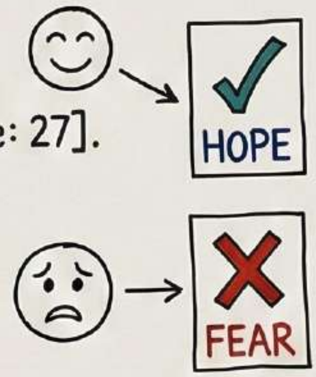
- Machine Learning techniques were used to analyze user behavior [cite: 19] and build **psychographic profiles** [cite: 20].

## Psychographic Profiling (continued from previous page)

- Categorized users based on: **personality traits** [cite: 22], **fears** [cite: 23], political leanings [cite: 24], **emotional vulnerabilities** [cite: 25].

## Micro-Targeting Technique

- Cambridge Analytica applied **micro-targeting**, where different users saw different political messages [cite: 27].
- Examples:
  - Strong supporters → shown positive and triumphant visuals [cite: 29].
  - Swing voters → shown negative content about opposing candidates [cite: 30].
- Messages were **personalized**, emotionally persuasive, and difficult to detect publicly [cite: 31].



Micro-Targeting in Action

## Political Campaigns Involved

- Consultant for:
  - **Donald Trump's 2016 U.S. Presidential campaign** [cite: 34].
  - **Leave.EU Brexit campaign** [cite: 35].
- ML-driven targeting influenced: voter perception [cite: 37], political engagement [cite: 38], democratic decision-making [cite: 39].

Memory Trick: Think "Scale & Exploit" for ML risks.

## Ethical Issues Involved

- **Lack of Informed Consent**: Users did not agree to political use of their data [cite: 41, 42].
- **Violation of Privacy**: Personal data used beyond original purpose [cite: 43, 44].
- **Behavioral Manipulation**: Psychological targeting influenced user decisions [cite: 45, 46].
- **Loss of Autonomy**: Users unknowingly guided toward specific political beliefs [cite: 47, 48].
- **Unfair Democratic Influence**: Political power gained through opaque data practices [cite: 49, 50].

# * Ethical Evaluation (ML Perspective)

- Target system: (unethical) (political manipulation) [cite: 52].
- Process: (unethical) (covert data harvesting) [cite: 53].
- Data source: (unethical) (no valid consent) [cite: 54].
- Data use: (unethical) (repurposing personal data) [cite: 55].
- Impact: (unethical) (threat to democracy and trust) [cite: 56].

| | |
|---|---|
| Target | ✗ |
| Process | ✗ |
| Data Source | ✗ |
| Data Use | ✗ |
| Impact | ✗ |

**UNETHICAL**


# * Why This Case Is Important in ML Ethics

- Demonstrates the **power of data-driven ML systems** [cite: 58].
- Shows how ML can:
  - **scale manipulation** [cite: 60].
  - **exploit psychological vulnerabilities** [cite: 61].
- Highlights risks when:
  - **transparency is absent** [cite: 63].
  - **accountability is avoided** [cite: 64].
  - **profit** or **political gain overrides ethics** [cite: 65].